

- (a)** To produce a human protein for treatment of a disease, recombinant DNA technology needs a gene coding for the particular human protein.

[4]

- Before recombinant human insulin became available, animals were the main source of insulin.

..... [3]



- (c) Explain why the DNA involved in the production of recombinant human insulin is termed **recombinant** DNA.

.....
..... [1]

- (d) Recombinant human insulin analogues are insulin proteins that have slightly altered amino acid sequences compared with recombinant human insulin. These analogues can be more effective than human insulin.

Synthetic genes coding for insulin analogues have been developed. The bacterium *Escherichia coli* can be used as a host for a synthetic gene for the large-scale manufacture of an analogue.

When scientists have determined the changes that are needed to produce an insulin analogue, they can obtain a synthetic gene coding for the analogue by making changes to a length of DNA using genetic engineering.

Suggest how scientists genetically engineer a synthetic gene coding for the insulin analogue **and** explain how the changes they make allow the correct analogue to be produced.

.....
.....
.....
.....
.....
.....
.....
.....
.....
..... [4]

[Total: 12]